

Automated Attendance Management System

Project Name: Automated Attendance Management System (AAMS)

Document Type: Business Requirements & System Analysis

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Date: May 2025

Introduction & Executive Summary

Executive Summary & Problem Statement

The current manual, paper-based attendance tracking system presents significant operational inefficiencies. It is time-consuming for academic staff, highly susceptible to fraudulent entries (proxy attendance), and incurs unnecessary printing and administrative costs. Furthermore, the lack of a centralized, real-time database prevents students and administration from instantly accessing attendance records, leading to communication gaps and disputes. As the student population continues to grow, scaling this manual process has become increasingly unsustainable.

Proposed Solution & Business Value

To address these critical challenges, this project proposes the development and implementation of a secure, electronic Automated Attendance Management System. By leveraging biometric verification and encrypted daily access codes linked to a centralized database, the proposed system will eliminate paper waste, enforce strict attendance integrity, and provide real-time reporting dashboards for administration, professors, and students.

This digital transformation aims to optimize operational workflows, significantly enhance campus security, and deliver a projected annual cost saving of 200,000 EGP.

Survey Phase

Request for system service

Request For Information Services	
Date of Request : 9/04/2025	Service Requested Departments: IT Department, Faculty Administration
Submitted By : Name : Ahmed Mohamed Role : CS Department Student	
Type of Services Requested: ☒ Business Process Analysis and Redesign ☐ Existing Application Enhancement ☒ New Applications Development ☐ Existing Application Maintenance	
Brief Statement of Problems, Opportunities, or Services: Paper-based attendance is slow, error-prone, and costly, students cheat, no real-time records. Need an electronic system with gates, encrypted codes, location verification.	
Brief Statement of Expected Results: Automate attendance, stop cheating, enable real-time record access, save costs, improve security.	
Assigned to: IT Department	Authorized Signature: Ahmed Mohamed Student, Computer Science Department
Approved Budget: 200,000 EGP	Start Date: May , 1 ,2025 Deadline: Oct 1 , 2025

Problem Statement

Problem Statement			
Project: Electronic Attendance System		Project Manager: Ahmed Mohamed	
Created By: Ahmed Mohamed		Last Updated By: Ahmed Mohamed	
Date Created: April 7, 2025		Date Last Updated: April 10, 2025	
Problem/Opportunities/Directives	Priority	Annual Benefits (EGP/Year)	Proposed Solution (in brief)
1. The paper-based attendance system is slow, error-prone, and costly; students cheat by marking friends as present, with no real-time records, causing disputes.	1	120,000 EGP	Automate with an electronic system using entry gates, encrypted codes, and location verification.
2. Students can't check records or get notifications, leading to confusion.	2	80,000 EGP	Provide real-time access and notifications via a college app.
1. Slow, error-prone, costly system, cheating, no real-time records: 120,000 EGP, 60% of total benefits. 2. No real-time access or notifications, causing confusion: 80,000 EGP, 40% of total benefits. Total Annual Benefits: 200,000 EGP			

Scope Statement

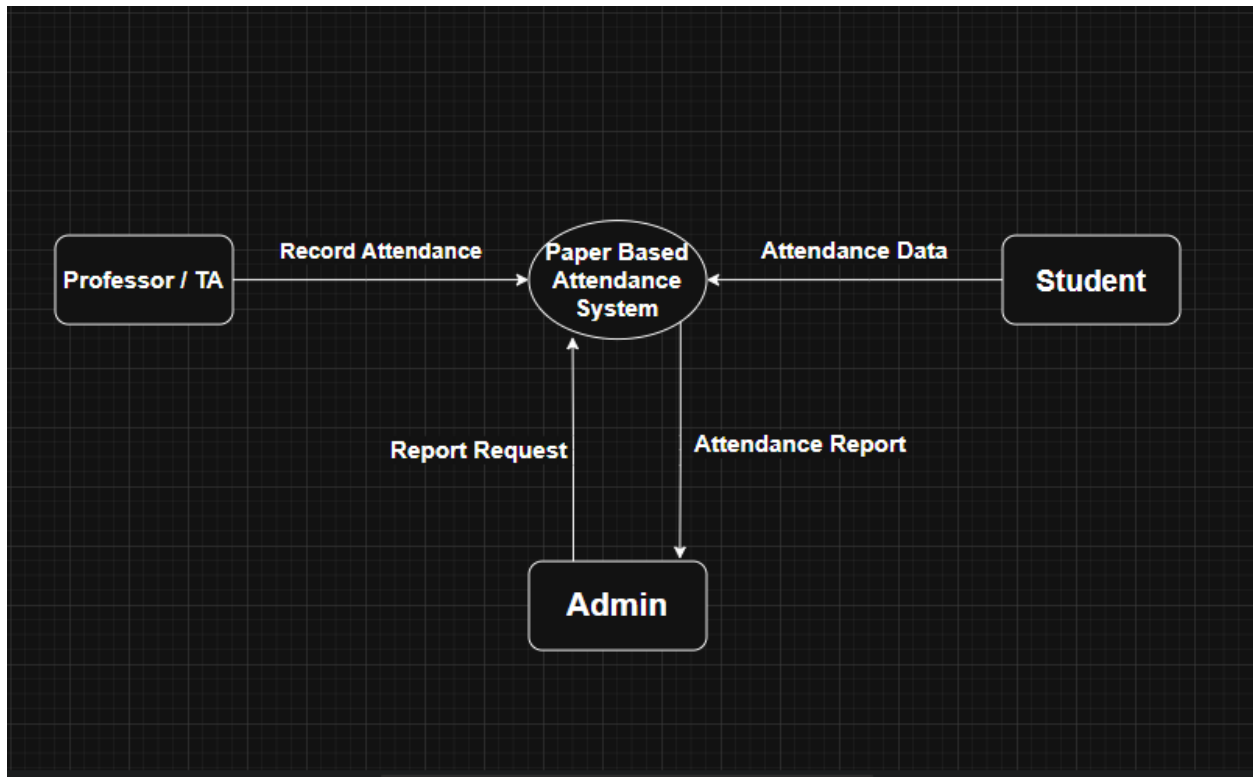
Representative Scope Statement				
Project : Electronic Attendance System		Project Manager: Ahmed Mohamed		
Created By: Ahmed Mohamed		Last Updated By: Ahmed Mohamed		
Date Created: April 7, 2025		Date Last Updated: April 10, 2025		
Brief Statement of Problems, Opportunities, or Directives	Data	Process	Expecting Time	Expecting Cost
1. Install entry gates at the university's main entrance to generate encrypted daily codes for students	Student IDs, Biometric Data, Encrypted Codes	Verify entry, generate codes	2 months	80,000 EGP
2. Develop a college app for students to record attendance with location verification and single-use policy Record	Attendance Records, Location Data	attendance, verify location	2 months	60,000 EGP
3. Enable real-time access and notifications for students via the app	Attendance Records, Notifications	Provide access, send notifications	15 days	20,000 EGP
4. Allow Admin and Professors/TAs to generate reports from a central	Attendance Reports, Database	database Generate reports, store data	15 days	40,000 EGP
			5 months	200,000 EGP

Master Plan

No	Task Name	May				Jun				Jul				Aug				Sept			
1	Survey Phase																				
2	Study Phase																				
3	Definition Phase																				
4	Configuration Phase																				
5	Design Phase																				
6	Procurement Phase																				
7	Implementation Phase																				
8	Delivery Phase																				
9	Fact-finding																				
10	Documentation																				
11	Presentation																				
12	Project Mgt.																				

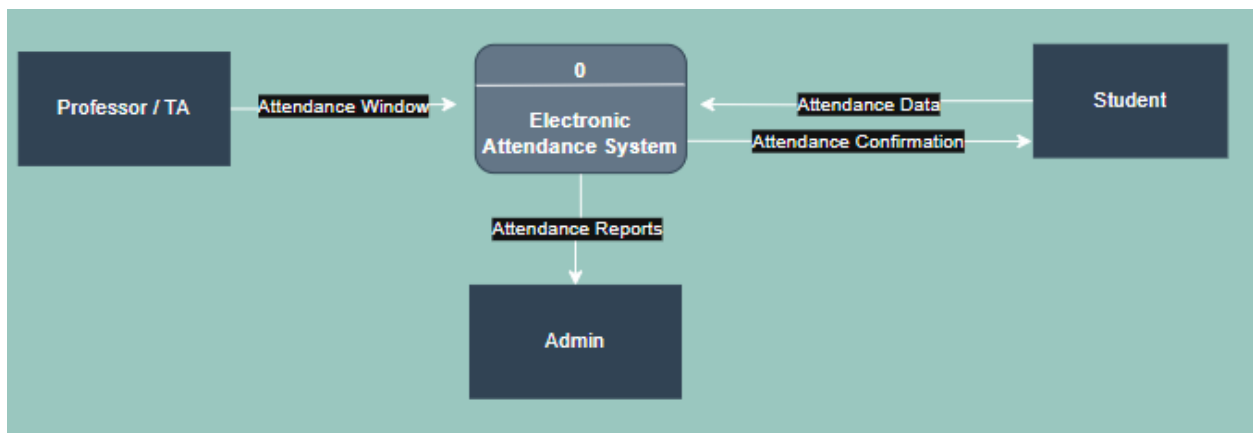
Study Phase

Current Data Flow Diagram (DFD) of the Current System (System As Is)

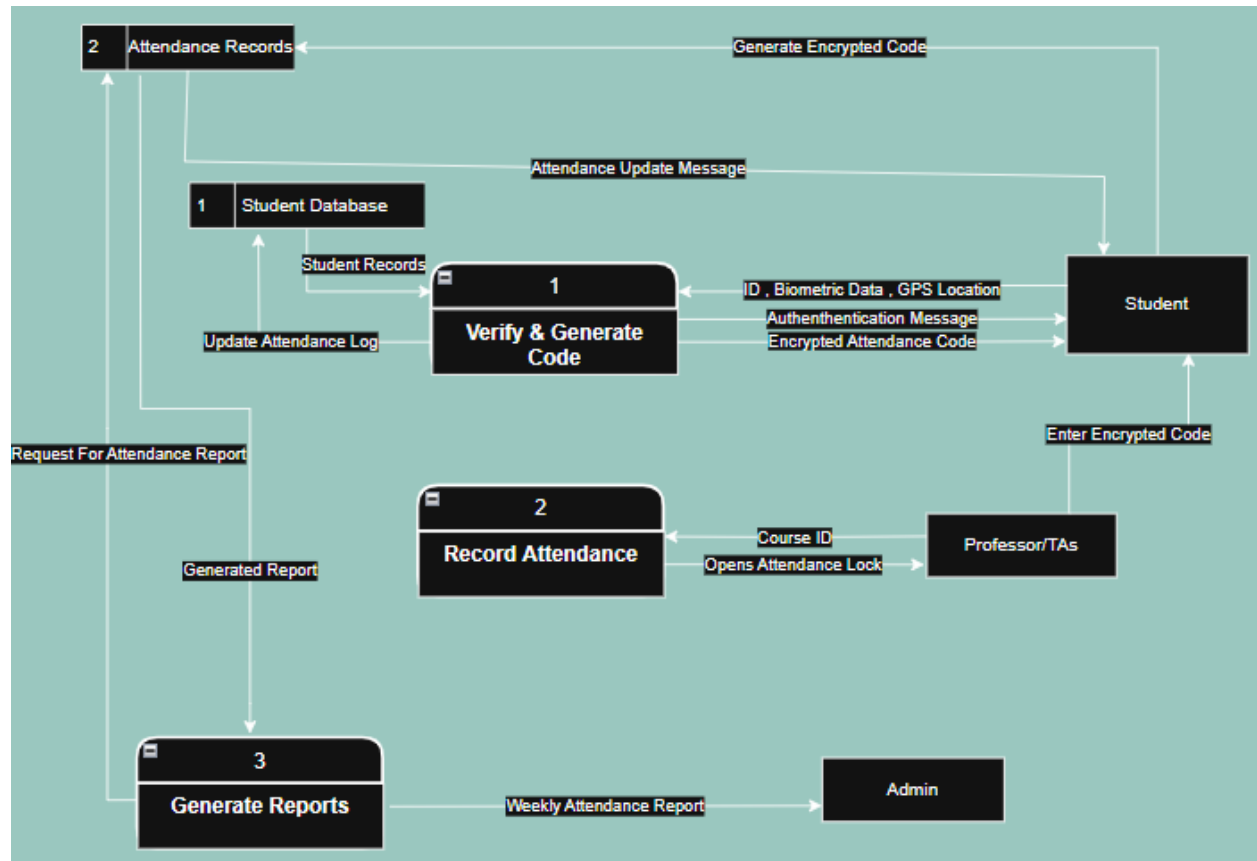


Purposed System (System To Be)

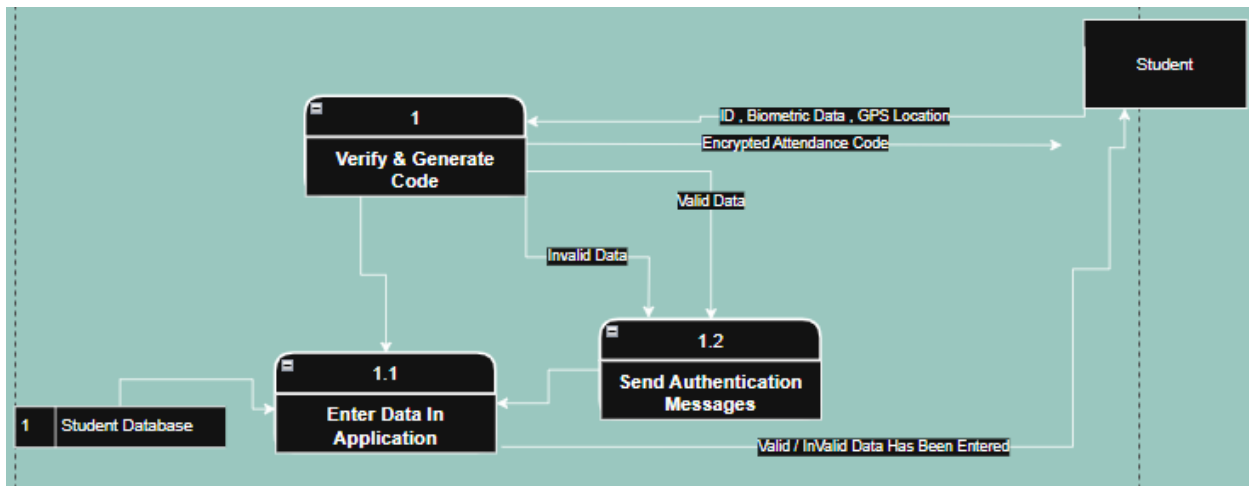
1. Context Level DFD



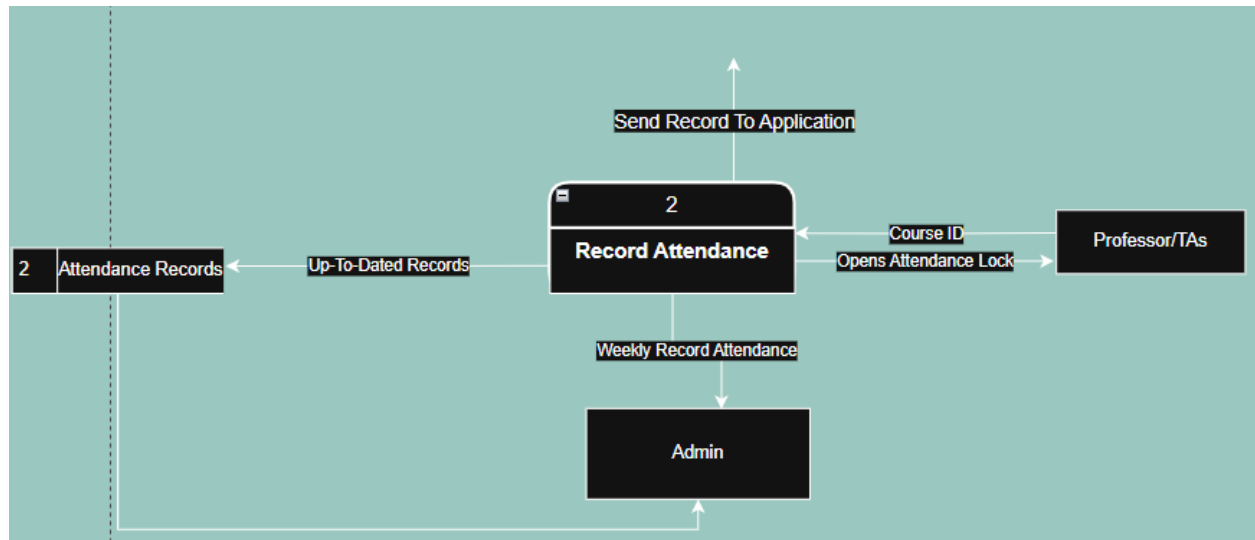
1- DFD Level-0



3- DFD Level-1 (More detailed breakdowns of internal processes)



4- DFD Level-2 (More detailed breakdowns of internal processes)



Definition Phase

Process Description Form

Process Name:	Verify Entry and Generate Code
Inputs:	- Entry Data (ID, Biometric) (from Student)
Outputs:	- Encrypted Code Generated (to Student) - Check Student Data (to Student Database) - Matched Data (from Student Database)
Description:	The process receives the student's ID and biometric data (fingerprint) at the university's entry gate. It verifies the data by checking it against the Student Database. If the data matches, it generates a daily encrypted code and sends it to the student for use in attendance recording. The process ensures secure verification to prevent unauthorized access
Dependencies:	- Student must provide valid ID and biometric data. - Student Database must be up-to-date with student records.
Exceptions:	- Invalid ID or biometric: Access denied, error message sent to student. - Database unavailable: Verification fails, manual check required - Multiple entry attempts: Only one code/Day

Data Store Description Form

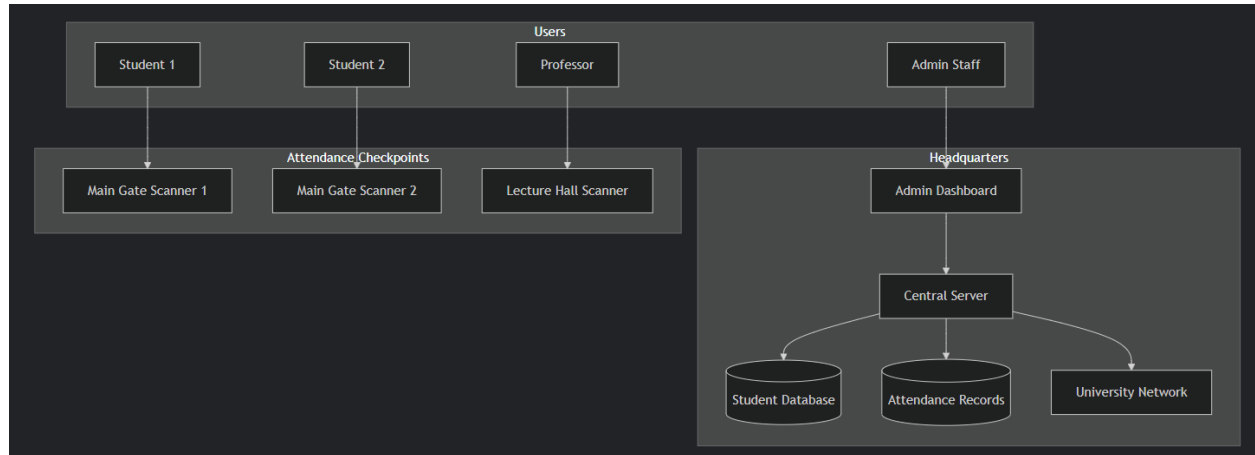
Data Store Name :	Student Data Base
Description :	Stores All Student Related information Including ID , Biometric data
Dependencies :	-Student ID -Student Biometric Data -Student Name -Enrollment Statue (To Deny Access IF the Enrollment
Inputs :	-Student Data -Check Data For The Verifying Gates
Outputs :	-Return True if the Entered Data Is Matched
Access :	-Process 1 (Read) -Entity 1 (Write)

Data Flow Description Form

Data Flow Name:	Entry Data (ID , Biometric)
Source :	Student
Destination :	Process 1 - Verify Entry and Generate Code
Description :	Contains Student's ID and Biometric Data
Frequency :	Once Per Operational Day
Constrains :	-Must Be Matched With Student's Database -ID AND Biometric Data Must Be Matched -Must Occur During Operational Day And Operational Hours

System Design Phase (Phase 2)

1- STA for Automated Attendance System



2- Technical Architecture Decisions

Question	Decision
Will the system use centralized or distributed computing?	Centralized
Will the system's data stores be centralized or distributed?	Centralized
What data storage technology will be used?	MySQL Database
Will the software be purchased or built in-house?	Built in-house
What programming language will be used?	C#
How will users interface with the system?	Hybrid: Mobile App + Web Dashboard
How will data be entered into the system?	Automated (Biometric Scan)
How will output be generated?	Real-time Screen Display + PDF/Excel
Will the system connect to old (legacy) systems?	No legacy systems connected

Input , Output & Interface

Electronic Attendance Management System

FileManagementToolsHelp

Student Verification

Student ID:

Biometric Data:

Scan Fingerprint

Generated Code:

Verify Generate Code

Record Attendance

Select Course:

Attendance Code:

Location Status:

Pending Verification...

Submit Attendance

Attendance Reports

	Student ID	Student Name	Course	Date	Time	Status
▶	S12345	Ahmed Moh...	CS101	2025-05-08	09:15 AM	Present
	S12346	Sara Ahmed	CS101	2025-05-08	09:18 AM	Present
	S12347	Mohamed Ali	CS101	2025-05-08	---	Absent

Filter by Course:

All Courses

Apply

Export Report

Electronic Attendance Management System

File

Management

Tools

Help

Student Verification

Student ID:

S12345

Biometric Data:

Scan Fingerprint

Generated Code:

ABCD2505080124

Verify Generate Code

Record Attendance

Select Course:

Attendance Code:

Location Status:

Pending Verification...

Submit Attendance

Attendance Reports

Student ID

Student Name

Course

Date

Time

Status

S12345

Ahmed Mohamed

CS101

2025-05-08

08:15 AM

Present

S12346

Sara Ahmed

08:18 AM

Present

S12347

Mohamed Ali

Absent

Filter by Course:

All Courses

Apply

Export Report

Student verified successfully! Code generated.

OK

Ready 2025-05-08 01:24:43

Electronic Attendance Management System

File

Management

Tools

Help

Student Verification

Student ID:

S12345

Biometric Data:

Scan Fingerprint

Generated Code:

ABCD2505080124

Verify Generate Code

Record Attendance

Select Course:

CS202 - Data Structures

Attendance Code:

ABCD2505080124

Location Status:

Verified (University Campus)

Submit Attendance

Attendance Reports

Student ID

Student Name

Time

Status

S12345

Ahmed Moh...

09:15 AM

Present

S12346

Sara Ahmed

09:18 AM

Present

S12347

Mohamed Ali

Absent

CS101

2025-05-08

Filter by Course:

All Courses

Apply

Export Report

Attendance recorded successfully!

OK

Student Verification

Student ID: Biometric Data: Generated Code:

Record Attendance

Select Course: Attendance Code: Location Status: Verified (University Campus)

Attendance Reports

	Student ID	Student Name	Report exported successfully!		Time	Status
▶	S12345	Ahmed Moh...		OK	09:15 AM	Present
	S12346	Sara Ahmed			09:18 AM	Present
	S12347	Mohamed Ali	CS101		2025-05-08	---

Report exported successfully!

Filter by Course: